

TKINTER PROGRAMS – 5 EXAMPLES

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Program 1 – Simple Greeting Window

Aim: To develop a simple graphical user interface using Python Tkinter.

Program:

```
import tkinter as tk

root = tk.Tk()
root.title("Greeting")
root.geometry("400x250")

label = tk.Label(root, text="Welcome to Tkinter!", font=("Arial", 18))
label.pack(pady=70)

root.mainloop()
```

Result: The Tkinter application runs successfully and displays the required GUI.

Program 2 – Student Details Form

Aim: To develop a simple graphical user interface using Python Tkinter.

Program:

```
import tkinter as tk

def display():
    result.config(text="Name: " + name.get() + "\nCourse: " + course.get())

root = tk.Tk()
root.title("Student Details")
root.geometry("400x300")

tk.Label(root, text="Student Name").pack(pady=5)
name = tk.Entry(root)
name.pack()

tk.Label(root, text="Course").pack(pady=5)
course = tk.Entry(root)
course.pack()

tk.Button(root, text="Display", command=display).pack(pady=15)
result = tk.Label(root, text="")
result.pack()

root.mainloop()
```

Result: The Tkinter application runs successfully and displays the required GUI.

Program 3 – Simple Calculator

Aim: To develop a simple graphical user interface using Python Tkinter.

Program:

```
import tkinter as tk

def calculate():
    a = float(num1.get())
    b = float(num2.get())
    op = operation.get()

    if op == "+":
        ans = a + b
    elif op == "-":
        ans = a - b
    elif op == "*":
        ans = a * b
    elif op == "/":
        ans = a / b if b != 0 else "Cannot divide by zero"

    result.config(text="Result: " + str(ans))

root = tk.Tk()
root.title("Calculator")
root.geometry("400x350")

tk.Label(root, text="Number 1").pack(pady=5)
num1 = tk.Entry(root)
num1.pack()

tk.Label(root, text="Number 2").pack(pady=5)
num2 = tk.Entry(root)
num2.pack()

operation = tk.StringVar(value="+")
tk.OptionMenu(root, operation, "+", "-", "*", "/").pack(pady=10)

tk.Button(root, text="Calculate", command=calculate).pack()
result = tk.Label(root, text="", font=("Arial", 14))
result.pack(pady=15)

root.mainloop()
```

Result: The Tkinter application runs successfully and displays the required GUI.

Program 4 – Login Form

Aim: To develop a simple graphical user interface using Python Tkinter.

Program:

```
import tkinter as tk
from tkinter import messagebox

def login():
    if username.get() == "admin" and password.get() == "1234":
        messagebox.showinfo("Login", "Login Successful")
    else:
        messagebox.showerror("Login", "Invalid Username or Password")

root = tk.Tk()
root.title("Login Form")
root.geometry("400x300")

tk.Label(root, text="Username").pack(pady=8)
username = tk.Entry(root)
username.pack()

tk.Label(root, text="Password").pack(pady=8)
password = tk.Entry(root, show="*")
password.pack()

tk.Button(root, text="Login", command=login).pack(pady=20)

root.mainloop()
```

Result: The Tkinter application runs successfully and displays the required GUI.

Program 5 – Temperature Converter

Aim: To develop a simple graphical user interface using Python Tkinter.

Program:

```
import tkinter as tk

def convert():
    c = float(celsius.get())
    f = (c * 9/5) + 32
    result.config(text=f"Fahrenheit: {f:.2f}")

root = tk.Tk()
root.title("Temperature Converter")
root.geometry("400x250")

tk.Label(root, text="Enter Celsius").pack(pady=10)
celsius = tk.Entry(root)
celsius.pack()

tk.Button(root, text="Convert", command=convert).pack(pady=15)
result = tk.Label(root, text="", font=("Arial", 14))
result.pack()

root.mainloop()
```

Result: The Tkinter application runs successfully and displays the required GUI.